Topological characteristics of the street network in São Paulo City, Brazil

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Street networks are critical infrastructure systems that can be severely disrupted by natural or human-made events, making the analysis of their structural properties essential for urban planning and emergency management. This study investigates the topological characteristics of the street network of São Paulo, Brazil. Four graph-based metrics were computed: node degree (k_i), node clustering coefficient (c_i), node average shortest path length (l_i), and edge betweenness centrality (e_{ij}). Additionally, the network was compared to an equivalent random graph $G(N, p)$ to assess its structural properties. The network was modeled as a directed multigraph with 122,456 nodes and 304,027 edges. Results show that the degree distribution does not follow a power-law behavior, indicating the network is not scale-free, with most intersections having a degree value of 6. A small subset of edges concentrates high e_{ij} values, identifying critical street segments whose failure could significantly disrupt connectivity across the city. Compared to an equivalent random graph, the real network presents a substantially higher clustering coefficient and an average shortest path length approximately 18 times larger, reflecting the geographical constraints inherent to physical street infrastructure. These findings contribute to understanding the structural vulnerability of São Paulo's street network and provide a foundation for future resilience analyses.

Keywords. urban street network, complex networks, network topology, edge betweenness centrality, São Paulo

1 Introduction

A street network can be modeled as a graph. A graph is a mathematical structure used to model pairwise relations between entities, regardless of their nature, enabling the analysis of connectivity patterns, structural properties, and dynamic processes across a wide range of domains, from social interactions and biological systems to transportation and urban infrastructure [7].

Street networks are vulnerable to disasters, which can cause damage to the infrastructure and adversely affect travel on the degraded network. Events such as floods can affect street serviceability. After such events, certain places often become less accessible. Studying and analyzing the topological characteristics of street networks helps prioritize planning, budgeting, and maintenance, as well as preparing effective emergency response plans [2].

Although the literature on street network vulnerability has advanced in proposing metrics and methodologies [6, 8], it is not yet fully clear how these structural properties behave in specific urban contexts. Each city has its own characteristics of street design, intersection density, and mobility patterns, which can directly influence graph measures. Understanding how these properties manifest in the studied spatial context is essential to reveal potential particularities that may guide local policies for planning and managing street infrastructure.

In this context, the objective of this study is to analyze the topological characteristics of urban streets in São Paulo, Brazil. The paper includes the analysis of the street network as a graph, with the computation of degree, average shortest path length, clustering, and edge betweenness.

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2 Material and methods

2.1 Study area

The area investigated included the complete territory of São Paulo, Brazil. From the selected area, it was possible to load the street network graph from January 2026, using the osmnx Python library [4].

2.2 Graph analysis

The street network is represented by a graph G, where each intersection is a node and each street segment is an edge. Therefore, V is a set of vertices (or nodes) and E is a set of edges (or links) that connects pairs of nodes (or connects a node to itself): $G = (V, E)$. The graph loaded is a Multidigraph, meaning that the graph can have multiple parallel edges between the same pair of vertices, possibly with distinct directions. This is a common feature of street networks, where bidirectional or unidirectional connections can exist between the same two intersections.

The edge (e) betweenness centrality (e_{ij}) is calculated by the sum of the fraction of all-pairs shortest paths that pass through that edge, as the following Equation shows:

$$e_{ij} = \sum_{s,t \in V} \frac{\sigma(s,t|e)}{\sigma(s,t)}; \tag{1}$$

where V is the set of nodes, $\sigma(s,t)$ is the number of shortest (s,t)-paths, and $\sigma(s,t|e)$ is the number of those paths passing through edge e [5].

Additionally, the node degree (k_i) was calculated, which is the number of edges connected to that node. This is known as adjacency (a_{ij}). When two nodes i and j are said to be neighbors, then $a_{ij} \neq 0$. The following Equation 2 shows the computation of k_i [5]:

$$k_i = \sum_j a_{ij} \tag{2}$$

The clustering coefficient (c_i) of a node i quantifies the proportion of existing connections among its neighbors relative to the maximum possible number of such connections [5]:

$$c_i = \frac{2L_i}{k_i(k_i - 1)} \tag{3}$$

where L_i is the number of edges between the neighbors of node i. The average shortest path length (l_i) of a node i is the mean of the shortest distances from i to all other reachable nodes [5]:

$$l_i = \frac{1}{N-1} \sum_{j \neq i} d_{ij} \tag{4}$$

where d_{ij} is the shortest path length between nodes i and j, and N is the total number of nodes.

Average values of degree ($\langle k \rangle$), clustering coefficient ($\langle c \rangle$) and average shortest path length ($\langle l \rangle$) were calculated and compared to a random network following the $G(N, p)$ model, where N is the quantity of nodes and p is the probability that these nodes are connected [3].

The probability p of this random graph can be calculated based on the number of edges L and the number of nodes N, with the following equation:

$$p = \frac{2L}{N(N-1)} \tag{5}$$

With a known value of average degree (which is the same as the average degree of the real network), it is possible to calculate clustering coefficient ($\langle c \rangle$) and average shortest path length ($\langle l \rangle$) for the random graph, with the following equations:

$$\langle c \rangle = \frac{\langle k \rangle}{N}; \langle l \rangle = \frac{logN}{log\langle k \rangle} \tag{6}$$

All the parameters were calculated using the NetworKit Python library [1]. All the calculations but one were performed on a MultiDigraph representation of the street network. The exception was the calculation of clustering coefficient, which was performed on a simple graph representation. Additionally, the random graph was generated using a simple graph representation. The code used for these calculations is available in the GitHub repository [9].

3 Results

The loaded graph resulted in 122,456 nodes and 304,027 edges (178,490 if only undirected edges are considered), representing the full extent of street network in São Paulo. Table 1 shows the distribution of the calculated parameters of the network. Degree values (k_i) vary between 1 and 11, with an average value of 4.9655. These values represent the behavior of the streets that are connected to the intersections, where some can be a two-way street or a one-way street. Both e_{ij} and c_i have median and minimum values of zero, and l_i also has a minimum value of 0 ($n = 65$), showing that these nodes cannot reach other nodes in a digraph.

Table 1: Distribution of network parameters.

Parameter	Mean	Median	Std	Min	Max
k_i	4.9655	6.0000	1.7093	1.0000	11.0000
c_i	0.0596	0.0000	0.1272	0.0000	1.0000
l_i	132.4250	126.0473	29.8825	0.0000	266.0343
e_{ij}	0.0004	0.0000	0.0041	0.0000	0.2029

In Figure 1, it is possible to observe the proportional distribution of k_i values. The São Paulo network, similar to most urban street networks, does not present a power-law behavior in the k_i distribution, and thus is not a scale-free network. Most of the nodes have a degree value of 6.

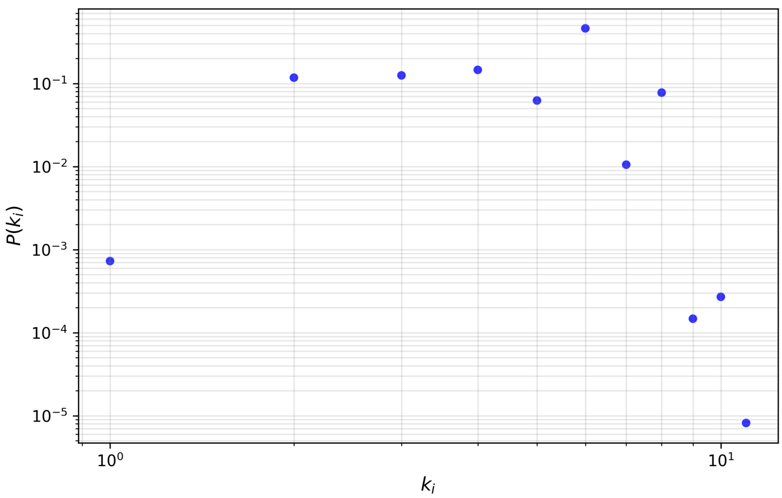

Figure 1: Distribution of k_i values. Source: The Authors (2026).

The map in Figure 2 shows the distribution of e_{ij} values across the network. It is observed that few edges have high values of e_{ij}, indicating street sections that are critical routes, likely carrying high traffic volumes and concentrating flow from many origin-destination pairs. It could also indicate locations of higher vulnerability to disruptions, impacting the overall network resilience and performance.

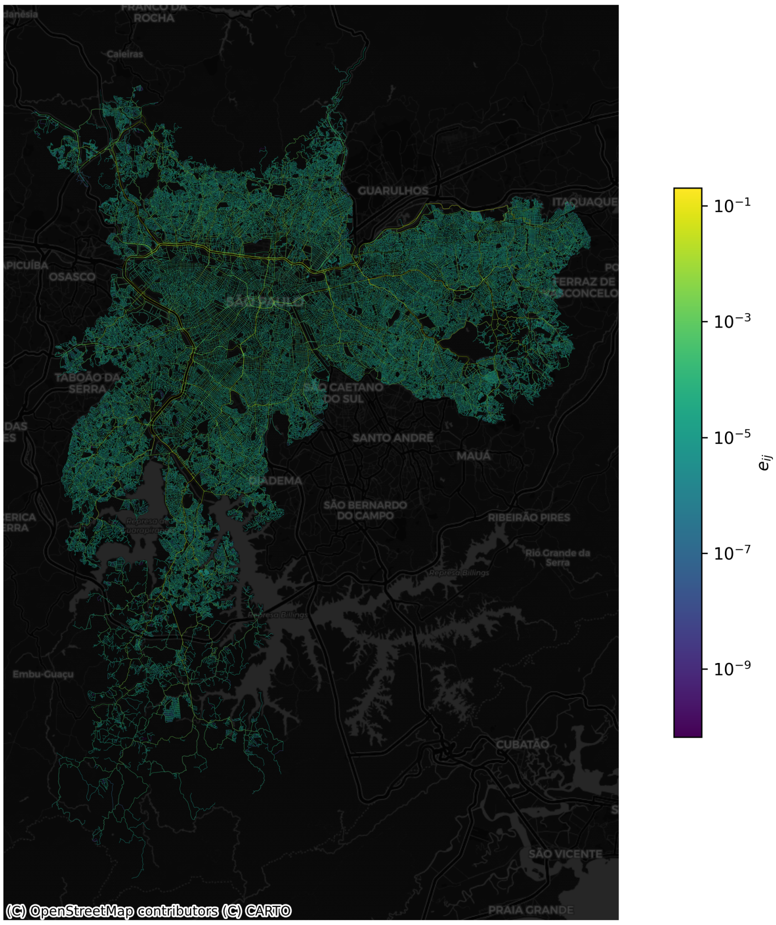

Figure 2: Map of e_{ij} results. Source: The Authors (2026).

In Table 2, it is possible to observe the comparison between the real São Paulo network and a theoretical random graph $G(N, p)$ to highlight the differences in their structural properties. The street network shows a higher value of $\langle c \rangle$ compared to the random graph, indicating that the real network has a strong local structure, which can be an attribute of how intersections are connected in the city (*e.g.* if intersection A connects to B and to C, there is a good chance B and C are also connected by a street, especially in grid-like urban layouts.)

Table 2: Comparison between the real network and the theoretical random graph $G(N, p)$.

Network	N	L	p	$\langle k \rangle$	$\langle c \rangle$	$\langle l \rangle$
Real network	122456	304027	–	4.9655	0.059600	132.4250
$G(N, p)$	122456	–	0.000041	4.9655	0.000041	7.3107

When comparing the $\langle l \rangle$ values, it is possible to observe that the real network has an 18 times higher average path length compared to the random graph. In the random graph, any node can connect to any other, creating long-range "shortcuts" that shorten distances. A real street network is constrained by geographical distances; therefore, to go from one side of the city to the other, a street user must traverse many intermediate edges (street segments) sequentially.

4 Conclusion

In this work, it was possible to analyze the topological properties of the São Paulo street network. The results showed that the network distribution of degree values did not follow a power-law behavior, indicating that the degree distribution is not scale-free. Regarding edge betweenness centrality, the network presented a few edges with higher values, representing street segments that are crucial for connecting different parts of the city.

When comparing the São Paulo street network with a theoretical random graph, the results highlight the differences in their structural properties. The real network shows a higher clustering coefficient and much longer paths (l_i), when compared to the random graph.

Future work may explore the computation and analysis of the Vulnerability Index on edges, to better understand the robustness of the network against failures. In this work, it was not possible to compute the Vulnerability Index due to the large size of the network and the computational resources required.

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